# Jorge Loría

*PhD in Statistics*

Jorge Loría /loriaJ jorge.loria@aalto.fi
/jorge-loría loriaj.github.io/

## Academic Employment

**Aalto University** June 2024–Present
*Postdoctoral Researcher, Advisor Dr. Samuel Kaski* *Espoo, Finland*

## Education

**Purdue University** 2019 – 2024
*PhD Statistics, Advisor: Dr. Anindya Bhadra* *Indiana, USA*
Thesis title: *On the use of $\alpha$-stable random variables in Bayesian bridge regression, neural networks and kernel processes* [link]

**Universidad de Costa Rica** 2015 – 2019
*BS Pure Mathematics* *San José, Costa Rica*

**Universidad de Costa Rica** 2014 – 2017
*BS Actuarial Science* *San José, Costa Rica*

## Research Interests

- Uncertainty Quantification
- Causal Inference

**Applications:**

- Bayesian inference
- Gaussian processes
- Personalized Medicine
- Computational methods
- Heavy-tailed distributions
- Environmental Sciences

## Publications

### Preprints

» Fernández-Loría, C. and **Loría, J.** (2026+), Inferring Effect Ordering Without Causal Effect Estimation *(submitted)* [arxiv]

### Publications

» Mäkinen L., **Loría, J.**, and Kaski, S. (2026), Bayesian Meta-Learning with Expert Feedback for Task-Shift Adaptation through Causal Embeddings *Proceedings of the 43rd International Conference on Machine Learning*, in *Proceedings of Machine Learning Research* 306:85894-85924 [link][github]

» Björkman, Z., **Loría, J.**, Wharrie, S., and Kaski, S. (2026), Incorporating Expert Knowledge into Bayesian Causal Discovery of Mixtures of Directed Acyclic Graphs. *Proceedings of The 29th International Conference on Artificial Intelligence and Statistics*, in **PMLR** 300:964-972 [link][github]

» **Loría, J.** and Bhadra, A. (2025), Deep Kernel Posterior Learning under Infinite Variance Prior Weights. *The 13th International Conference on Learning Representations (ICLR 2025).* [link] [github]

» **Loría, J.** and Bhadra A. (2024). Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance. *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, **PMLR 244** 2331-2349 [link] [github]

» **Loría, J.** and Bhadra, A. (2024). SURE-tuned bridge regression. *Statistics and Computing* **34**, 30. [doi link] [github]

» Víquez, J. J., Víquez, J., Campos, A., **Loría, J.**, Mendoza, L. A. (2018). Demographic Modeling via 3-dimensional Markov Chains *Revista de matemática Teoría y aplicaciones* available in https://revistas.ucr.ac.cr/index.php/matematica/article/view/33608

## Awards

**Somos Purdue Award** Class of 2024
recognition of commitment to excellence and for leading with *ganas* at Purdue University.

**Bravo Award** Spring 2024
one time staff award

**Ross-Lynn Research Scholar Grant** Spring 2024
provides funding for notable graduate students

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**Fully funded assistance to workshop on Bayesian Statistics and Statistical Learning** December 2023
IMSI, Illinois, USA

**Department of Statistics Travel Award** Summer 2023

**Department of Statistics Teaching Academy Associate Nominee** Spring 2023

**Emily and Paul Kidwell graduate student excellence Award** Spring 2022
given to graduate students in the Department of Statistics in Purdue University interested in careers in academia.

**Bob and Marjorie McLean Scholarship** Summers 2020 and 2021
given to distinguished graduate students of the Department of Statistics in Purdue University.

**Bronze Medal** 2015
awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas

**Honorable Mention** 2014
awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas

## Talks and Posters Presented

**Incorporando humanos en inferencia: descubrimiento causal y meta-aprendizaje, con aplicaciones médicas** October, 2026
*Invited talk in Instituto Tecnológico Autónomo de México (ITAM).* *Ciudad de México, México*

**Deep Kernel Posterior Learning under Infinite Variance Prior Weights** April, 2025
*Poster in International Conference on Learning Representations* *Singapore*

**Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance** July, 2024
*Poster in Uncertainty in Artifical Intelligence* *Barcelona, Spain*

**Predicción en redes neuronales Bayesianas infinitas con pesos de varianza infinita** May 22nd, 2024
*Seminario Intersede* *CIMAT, Guanajuato, México*

**SURE-tuned Bridge Regression** August 7th, 2023
*Contributed Paper in Joint Statistical Meeting* *Toronto, Canada*

**SURE-tuned Bridge Regression** July 10th, 2023
*Contributed talk in XVI Latin American Congress of Probability and Mathematical Statistics* *São Paulo, Brazil*

**SURE-tuned Bridge Regression** May 26th, 2023
*Poster in Statistics and Optimization in Data Science Workshop, in Purdue Business School* *Indiana, USA*

**SURE-tuned Bridge Regression** April 2023
*Invited talk in a weekly colloquium at CIMPA Universidad de Costa Rica.* *Online*

**Learning the Ranking of Causal Effects with Confounded Data** October 2022
*Flash talk in INFORMS* *Indiana, USA*

## Service to the Profession

**PhD. Thesis advisor**

» Lotta Mäkinen (ongoing, with S. Kaski as supervisor)

**MSc. Thesis advisor**

» Zachris Björkman (2024), Lotta Mäkinen (2025), and Axel Hedman (2025)

**Reviewer**

» Annals of Applied Statistics, AISTATS (2026), ICLR (2024, 2026), ICML (Silver Reviewer: 2026), Neurips (2023), TMLR (2026), Statistics in Medicine, UAI (2024, 2026)

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## Teaching Experience

**Aalto University** Fall 2022
*Instructor* *Espoo, Finland*

» Co-organized a seminar on causal inference, following Pearl's 'primer' book on causal inference.

**Purdue University** Fall 2022
*Instructor* *Indiana, United States*

» Taught one section of the course *Introduction to Statistics*. An introductory course for undergraduate students of computer science, mathematics, actuarial science, and statistics majors. Responsibilities included three weekly lectures, preparation of the relevant material, grading, and collaboration with the other instructors in the evaluation and design of the assignments of the course.

**Purdue University** Fall 2021 – Summer 2022, Summer 2023
*Instructor* *Indiana, United States*

» Taught two sections of the course *Elementary Statistical Methods*. An undergraduate level course for non-stem majors. Responsibilities included two weekly lectures to each section, preparation of the material, grading, leading teaching assistants and collaboration with other instructors for the evaluation of the course assignments.

**Purdue University** Fall 2019 – Spring 2021
*Teaching Assistant* *Indiana, United States*

» Held laboratories for two sections, office hours, and grading for the course *Elementary Statistical Methods*.

**Universidad de Costa Rica** Fall of 2017 & 2018
*Instructor* *San José, Costa Rica*

» Taught undergraduate course on basic functionalities of Excel and R. Responsibilities included lesson planning, course scheduling, and preparation of homework, exams, and course projects.

## Industry Experience

**Banco Promérica** January 2019 – July 2019
*Financial Risk Analyst, analyzed financial data to evaluate and forecast credit risk* *San José Costa Rica*

**Quanto Soluciones S.A.** January 2017 – July 2018
*Actuarial Model Programmer, implemented credit risk models* *San José Costa Rica*

## Communication

**Spanish** *Native* **English** *Advanced* **Portuguese** *Basic*

## Leadership Roles

**- President of Statistics Graduate Student Organization** 2021-2022
*Indiana, USA*

**- Vice-president of Costa Rican Students Association at Purdue (CRISaP)** 2021-2024
*Indiana, USA*